Algorithm (ECDSA) and Secure Hash Algorithm (SHA), employed to validate transactions and secure users' resources.

However, the emergence of quantum computers puts these cryptographic foundations at risk. Quantum computing algorithms, e.g., Shor's and Grover's, have the capability to disable classical encryption, making blockchain networks vulnerable to a breach. Hence, experts are currently searching for quantum-resistant cryptography solutions withstanding quantum attacks without sacrificing the decentralised aspect of blockchain.

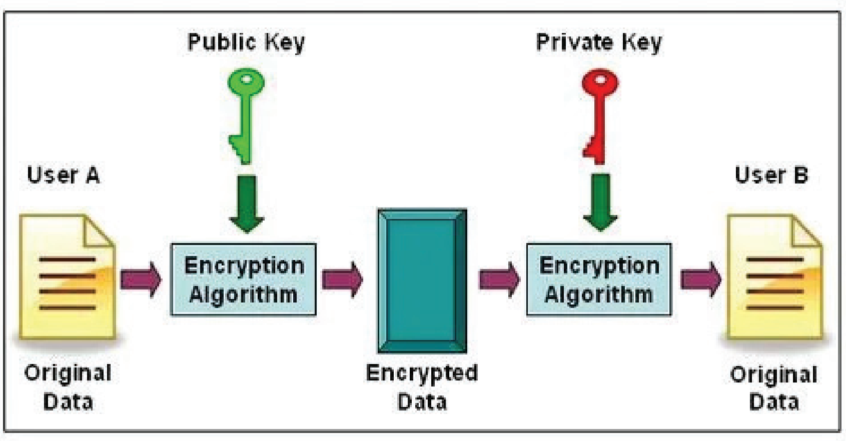

Figure 1: ECC (elliptic curve cryptography)

Quantum computing's challenge to blockchain security

Quantum computing uses the laws of quantum mechanics, like superposition and entanglement, to process information at exponentially greater speeds compared to conventional computers. This computing power, although beneficial in scientific pursuits, threatens blockchain security severely.

Shattering public-key cryptography: The greatest threat is from Shor's algorithm, which is capable of factoring large numbers and solving discrete logarithms. Both the mathematical underpinnings support RSA (Rivest–Shamir–Adleman) and ECC (elliptic curve cryptography)—two of the prevalent encryption algorithms employed in blockchain networks. The decryption private key generated from a public key can be deciphered by a sufficiently powerful quantum computer, making traditional blockchain security systems obsolete. That is, a quantum-capable attacker can take over digital assets, generate false transactions, and even manipulate blockchain consensus processes.

Collision attacks on hashing: Grover's algorithm also poses a serious threat. Compared to traditional

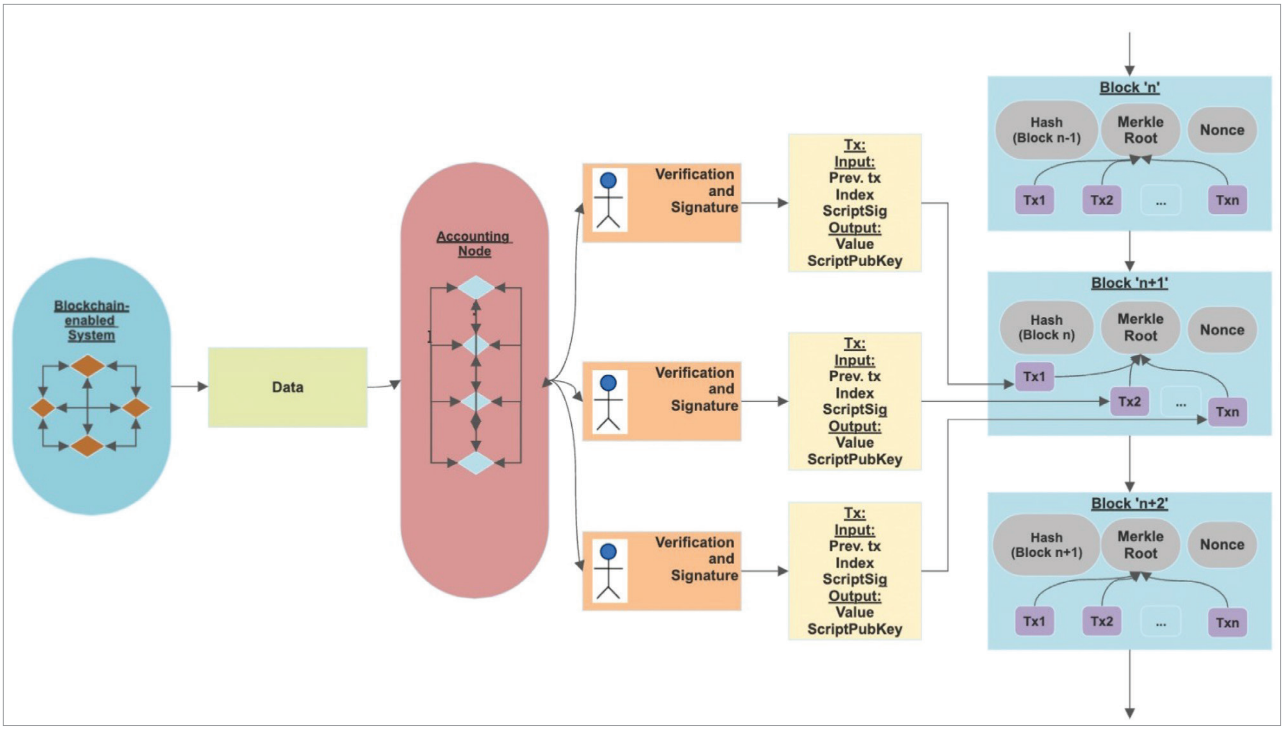

Figure 2: Quantum-resistant blockchain system